DSC 40B Discussion 6 Worksheet Solutions

Friday, May 8, 2026

Imagine you are a tutor, and you need to explain something to someone who has only a vague understanding of what's going on. Do your best to get the point across. **ALTERNATE**.

If you're not sure, ask your peer if they know how to explain it, then REPEAT the explanation back. Remember, the point of the exercise is to **vocalize** your thoughts.

If you are both unsure, make sure to come to/tutor for help! Do not use ChatGPT or any other LLM! The point of discussion is to prepare you for the exams.

Have one person in the pair open the Discussion 1 Gradescope assignment. After both you **and** your partner finish a problem **and explain it**, answer the corresponding multiple choice/short answer questions for both versions.. You will receive instant feedback on the questions, whether you are correct or not. If your answer is incorrect, please go back to the question to understand it and feel free to ask tutors about anything you are confused about. Once you have finished the entire worksheet or discussion is ending, turn in your assignment and add your partner to the submission.

Quick Icebreaker

Introduce yourself to your partner by sharing name, year, major, etc. What has been the biggest surprise this quarter? (Do not spend more than 2 minutes!)

Questions

Question 1:

First Person:

You are a city planner responsible for mapping out and building roads for cars. Your boss wants you to model out your roads in a graph. What kind of graph would you use to model it? Explain your reasoning.

- [ ] Undirected Graph
- [x] ~~Directed Graph~~

A directed graph would be more appropriate as roads inherently have a direction that cars should be going.

What would the vertices of your graph represent? Explain your reasoning.

There are multiple right answers here. Possible examples are intersections, buildings, turns in the road, or even cities from a more abstracted representation of roads/highway system.

Second Person:

Since you did so well planning out roads, your boss has promoted you to also be responsible for planning and building hiker trails for the local national park. Once again, your boss wants you to model out your trails in a graph. What kind of graph would you use to model it? Explain your reasoning.

- ☑ ~~Undirected Graph~~
- ☐ Directed Graph

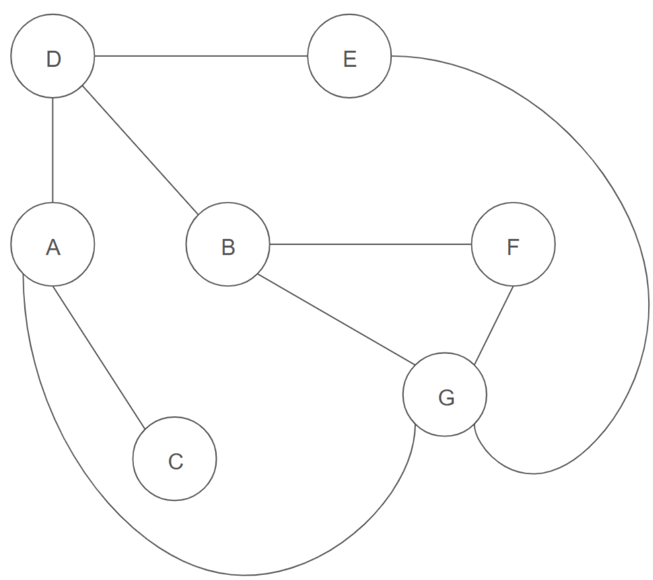
There is no specific direction for hiking trails (at least not that I know of). Hikers are able to walk in either direction, therefore modeling it as undirected edges would be more appropriate.

What would the vertices of your graph represent? Explain your reasoning.

Could be certain landmarks/viewing points, bathrooms, turns in the trail, intersections of trails.

Question 2:

Second Person:

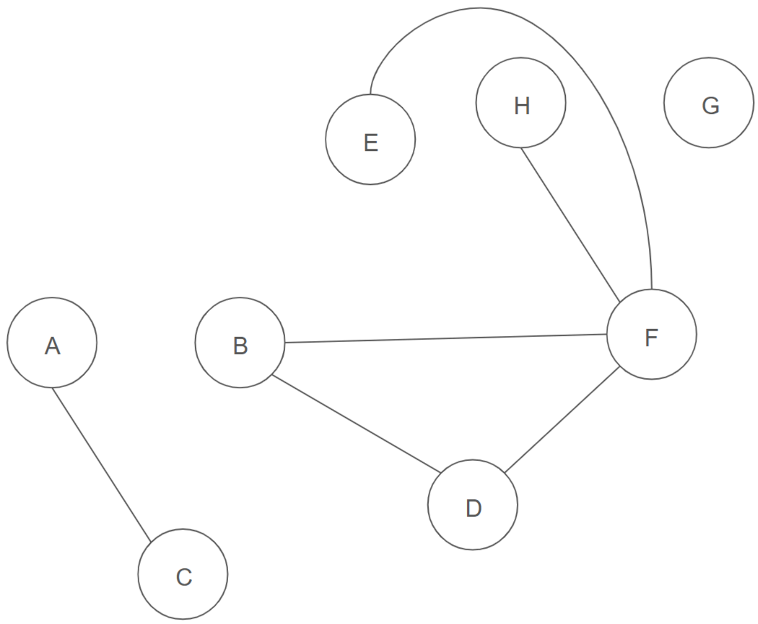


Given the graph above, list the set of vertices and edges.

{A, B, C, D, E, F, G}

{(A, C), (A, D), (A, G), (B, D), (B, F), (B, G), (D, E), (E, G), (F, G)}

First Person:



Given the graph above, list the set of vertices and edges.

{A, B, C, D, E, F, G, H}

{(A, C), (B, D), (B, F), (D, F), (E, F), (F, H)}

Question 3:

First Person:

Given that there are 11 vertices in a directed graph, what is the maximum number of edges possible?

121; Each vertex can have an edge directed at every other vertex and a self loop. Therefore each vertex has 11 outgoing edges, and with 11 total vertices, the maximum number of edges is 121.

What if the graph is undirected?

55; The maximum number of edges is 11 choose 2 = 55. To separate it out, you can say that the first vertex is connected to every other vertex, which gives 10 edges. The second vertex can be connected to every other vertex as well, which is 9 additional edges (as the connection with the first vertex is already considered). Therefore the maximum connections follows the pattern 10 + 9 + 8 + ... + 1 + 0, which is equivalent to n choose 2, where n is the number of vertices.

Second Person:

Given that there are 14 vertices in a directed graph, what is the maximum number of edges possible?

196; Each vertex can have an edge directed at every other vertex and a self loop. Therefore each vertex has 14 outgoing edges, and with 14 total vertices, the maximum number of edges is 196.

What if the graph is undirected?

91; The maximum number of edges is 14 choose 2 = 91. To separate it out, you can say that the first vertex is connected to every other vertex, which gives 13 edges. The second vertex can be connected to every other vertex as well, which is 12 additional edges (as the connection with the first vertex is already considered). Therefore the maximum connections follows the pattern 13 + 12 + 11 + ... + 1 + 0, which is equivalent to n choose 2, where n is the number of vertices.

Question 4:

Second Person:

Given that there are 14 edges in a directed graph, what is the minimum and maximum number of vertices possible?

The minimum is 4. A fully connected directed graph of 3 nodes has 9 edges. Therefore we must introduce a fourth node, where the graph can have a maximum of 16 edges, which is more than the 14 we must have.

The maximum is infinite. There can be an infinite number of nodes in a graph as there is no constraint that the graph must be fully connected.

What if the graph is undirected?

The minimum is 6. For a fully connected undirected graph of 5 nodes, the number of nodes is 10. Therefore we must introduce a 6th node, which would be 15 edges if it was fully connected.

The maximum is infinite. There can be an infinite number of nodes in a graph as there is no constraint that the graph must be fully connected.

First Person:

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Question 5:

First Person:

Given that an undirected graph has 8 vertices, what is the maximum degree a single node can have?

The maximum degree is 7. The node is connected to every other node in the graph, and since self loops cannot exist in an undirected graph, the maximum degree is n-1, where n is the total number of nodes.

What if the graph is directed?

The maximum degree is 16. The node has an outgoing edge to every other node in the graph and an incoming edge from every node. This would be a total of 14 edges. In addition, the self look counts as a degree of 2 since it is both an outgoing and incoming edge. Therefore the maximum degree is 16.

Second Person:

Given that an undirected graph has 9 vertices, what is the maximum degree a single node can have?

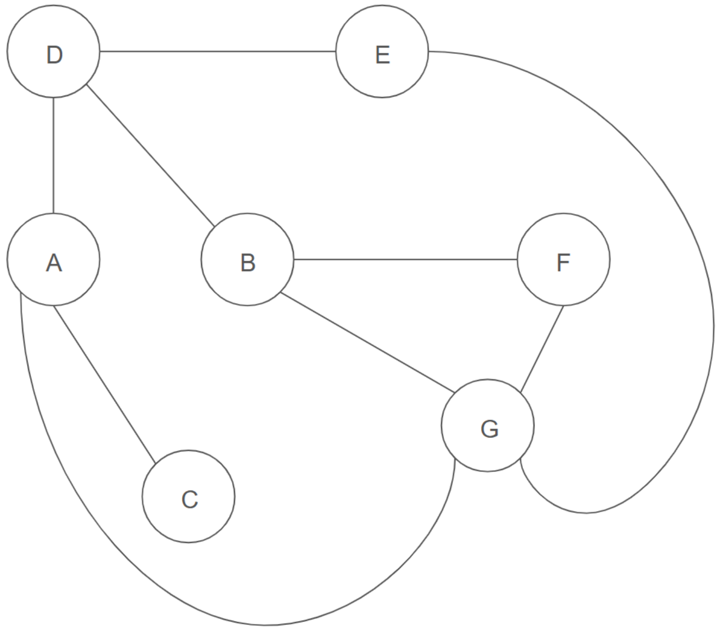
The maximum degree is 8. The node is connected to every other node in the graph, and since self loops cannot exist in an undirected graph, the maximum degree is n-1, where n is the total number of nodes.

What if the graph is directed?

The maximum degree is 18. The node has an outgoing edge to every other node in the graph and an incoming edge from every node. This would be a total of 16 edges. In addition, the self look counts as a degree of 2 since it is both an outgoing and incoming edge. Therefore the maximum degree is 18.

Question 6:

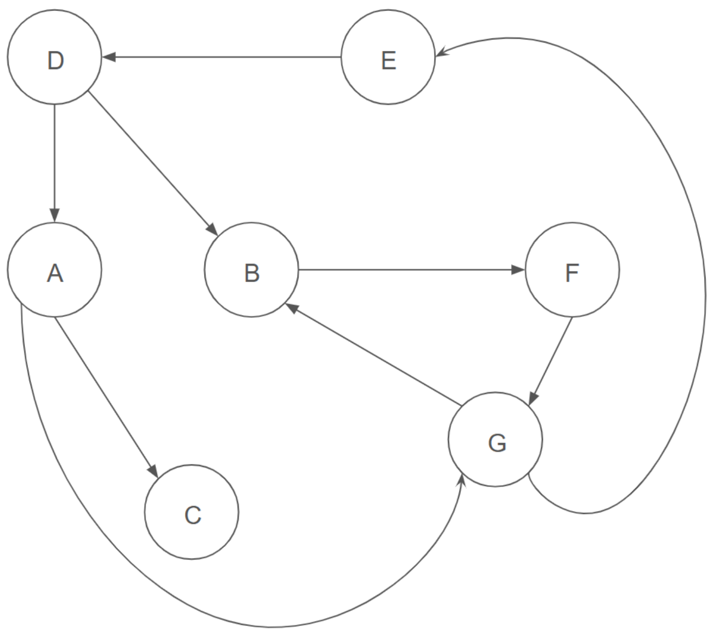
Second Person:



Give the graph above, list each node’s neighbors.

A: {C, D, G}
B: {D, F, G}
C: {A}
D: {A, B, E}
E: {D, G}
F: {B, G}
G: {A, B, E, F}

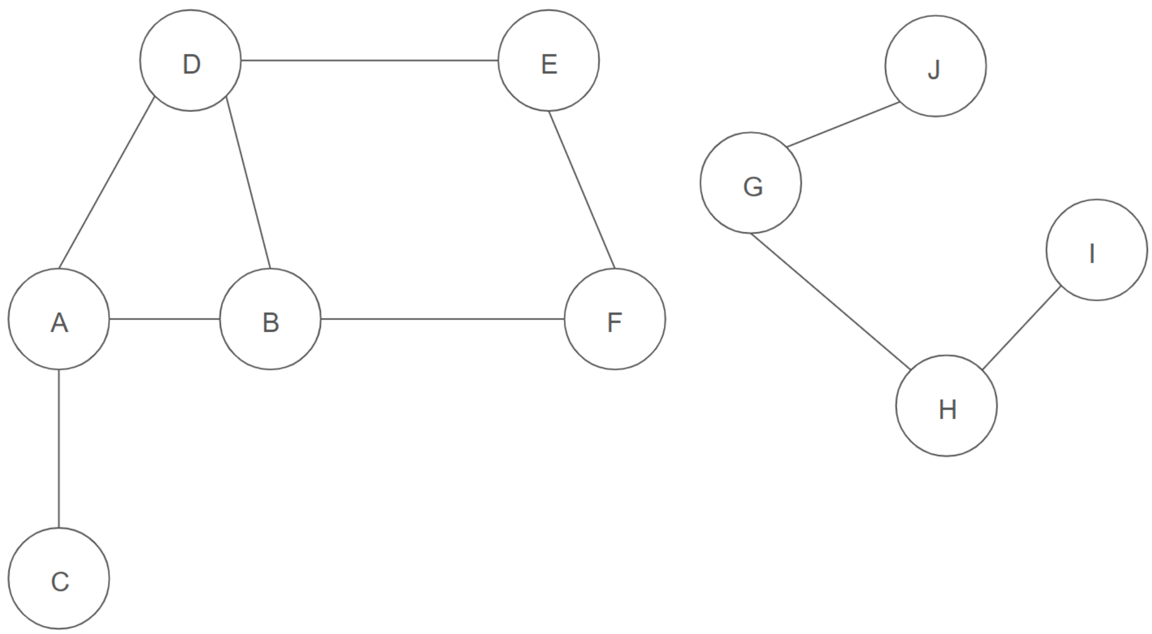
First Person:



Given the graph above, list each node’s neighbors.

A: {C, G}
B: {F}
C: {}
D: {A, B}
E: {D}
F: {G}
G: {B, E}

Question 7:



First Person:

Name a path between vertex J and I.

Many correct answers (ex. (J, G, H, I), (J, G, H, G, H, I), etc.)

What is the shortest path between vertex J and I?

The shortest path is J → G → H → I, which has a length of 3.

Second Person:

Name a path between vertex A and F?

Many correct answers (ex. (A, D, E, F), (A, B, F), etc.)

What is the shortest path between vertex A and F?

The shortest path is A → B → F, which has a length of 2.

Question 8:

Second Person:

What is the longest path between vertex A and F?

The longest path is of infinite length. There is no specification that the path must be a simple path. Therefore the path can go infinitely long.

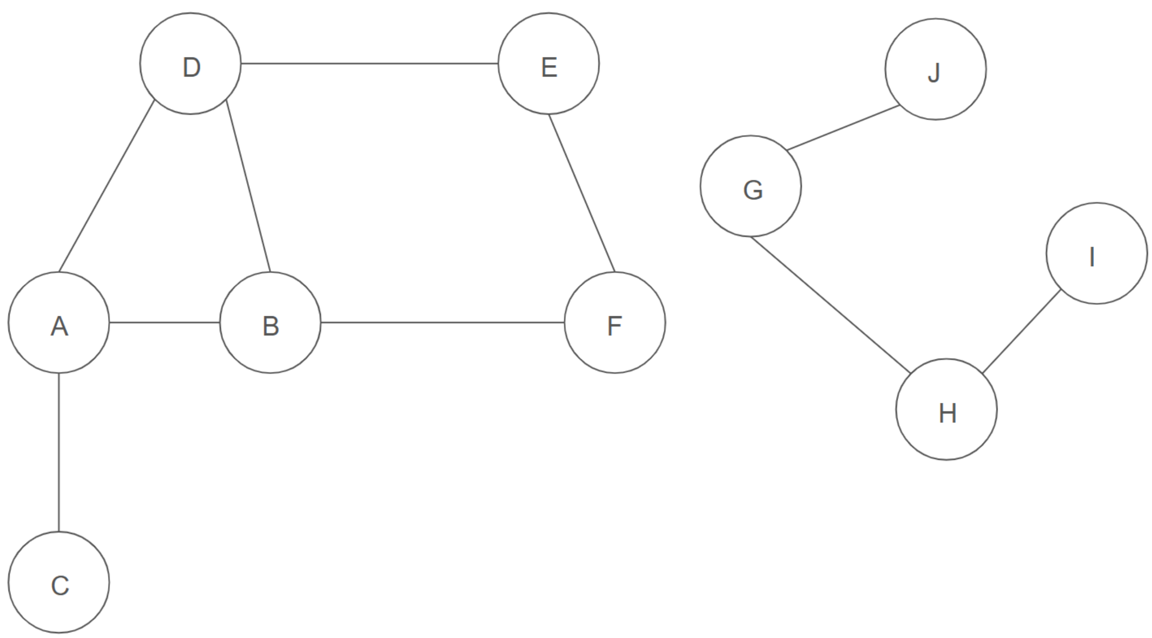
First Person:

What is the longest path between vertex J and I?

The longest path is of infinite length. There is no specification that the path must be a simple path. Therefore the path can go infinitely long.

Question 9:

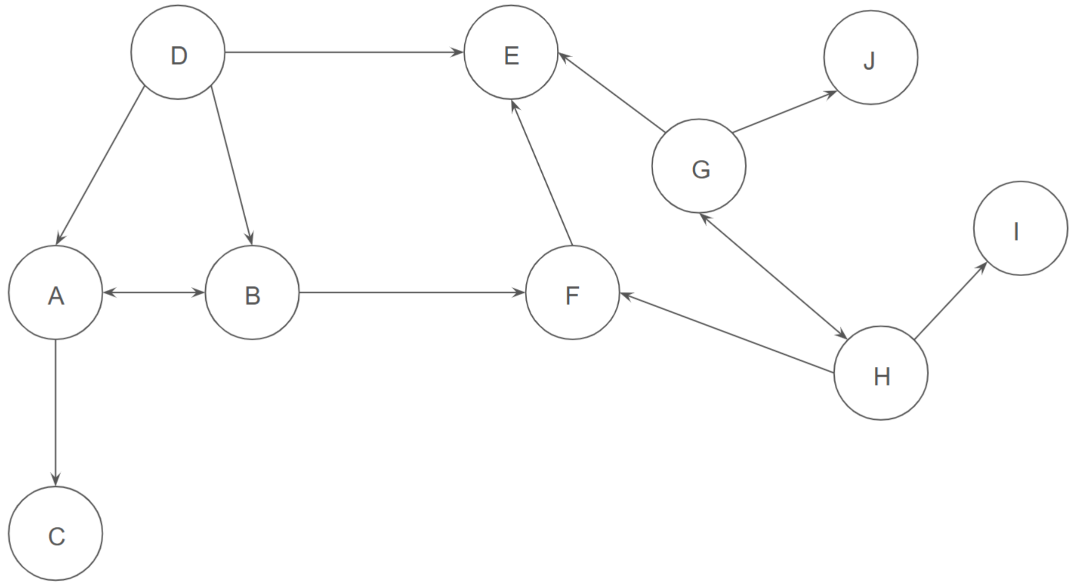
First Person:



Given the graph above, list all vertices reachable from vertex A.

{A, B, C, D, E, F}

Second Person:



Given the graph above, list all vertices reachable from vertex A.

{A, B, C, E, F}

Question 10:

Second Person:

For any connected **directed** graph, there exists a unique shortest path from each vertex to every other vertex in the graph.

- ☐ True
- ☑ ~~False~~

There can still be two different paths with the same shortest length between the two nodes.

First Person:

For any connected **undirected** graph, there exists a unique shortest path from each vertex to every other vertex in the graph.

- ☐ True
- ☑ ~~False~~

There can still be two different paths with the same shortest length between the two nodes.

Question 11:

First Person:

Given that an undirected graph has 8 vertices and 12 edges, what is the minimum and maximum number of connected components the graph can have?

The minimum number is 1. Can connect all nodes together to form one connected component.

The maximum number is 3. First form a fully connected component of 5 nodes using 10 edges (5 choose 2). Since there are 2 more edges, must connect an additional node to this component. Therefore we have one component with 6 nodes and 2 singular nodes not connected to any other node, giving us 3 total connected components.

Second Person:

Given that an undirected graph has 12 vertices and 8 edges, what is the minimum and maximum number of connected components the graph can have?

The minimum number is 4. Can at most connect 9 nodes together with 8 edges. Therefore we have 3 singular nodes along with the first connected component with 9 nodes, giving us 4 connected components.

The maximum number is 8. First form a fully connected component of 4 nodes using 6 edges (4 choose 2). Since there are 2 more edges, must connect an additional node to this component. Therefore we have one component with 5 nodes and 7 singular nodes not connected to any other node, giving us 8 total connected components.

Question 12:

Second Person:

If a graph is a binary tree, then its number of vertices is guaranteed to equal the number of edges plus one.

- [x] ~~True~~
- [ ] False

By definition of a tree, $|V|=|E|+1$, where $|V|$ is the number of vertices and $|E|$ is the number of edges.

First Person:

If a graph's number of vertices is equal to the number of edges plus one, then it is guaranteed to be a binary tree.

- [ ] True
- [x] ~~False~~

Can have three nodes all connected to each other (in a triangle) and another singular node by itself. This would not be a valid binary tree.

Aside: By definition of a tree, there must be no cycles.

Question 13:

First Person:

Consider an undirected graph with 5 nodes. Which two of the following choices represent the same graph?

A.
```
adj = [
    [1, 2],
    [0, 3],
    [0, 3, 4],
    [1, 2],
    [2]
]
```

B.
```
adj = [
    [1, 2],
    [0, 2, 3],
    [0, 1, 4],
    [1],
    [2]
]
```

C.
```
adj = [
    [1],
    [0, 3],
    [0, 3, 4],
    [1, 2],
    [2]
]
```

D.
```
adj_matrix = [
```

```
[0, 1, 1, 0, 0],
[1, 0, 0, 1, 0],
[1, 0, 0, 0, 1],
[0, 1, 0, 0, 0],
[0, 0, 1, 0, 0]
]
```

E.

```
adj_matrix = [
    [0, 1, 1, 0, 0],
    [1, 0, 1, 1, 0],
    [1, 1, 0, 0, 1],
    [0, 1, 0, 0, 0],
    [0, 0, 1, 0, 0]
]
```

F.

```
adj_matrix = [
    [0, 1, 1, 0, 0],
    [1, 0, 0, 1, 0],
    [1, 0, 0, 1, 1],
    [0, 1, 1, 0, 0],
    [0, 0, 1, 0, 0]
]
```

A, F

Second Person:

Consider a directed graph with 5 nodes. Which two of the following choices represent the same graph?

A.

```
adj = [
    [1, 3],
    [2],
    [4],
    [2],
    [3]
]
```

B.

```
adj = [
    [1],
    [2, 3],
    [0, 4],
    [2],
```

```
    [3]
]
```

C.

```
adj = [
    [1, 3],
    [2],
    [0, 4],
    [2],
    [3]
]
```

D.

```
adj_matrix = [
    [0, 1, 0, 1, 0],
    [0, 0, 1, 0, 0],
    [1, 0, 0, 0, 1],
    [0, 0, 1, 0, 0],
    [0, 0, 0, 1, 0]
]
```

E.

```
adj_matrix = [
    [0, 1, 0, 1, 0],
    [0, 0, 1, 0, 0],
    [1, 0, 0, 0, 1],
    [0, 1, 0, 0, 0],
    [0, 0, 0, 1, 0]
]
```

F.

```
adj_matrix = [
    [0, 1, 0, 1, 0],
    [0, 0, 1, 0, 0],
    [0, 0, 0, 0, 1],
    [0, 0, 1, 0, 0],
    [0, 0, 0, 1, 0]
]
```

C, D

Question 14:

You are running a large scale social network app with millions of users where users add each other as friends.

Second Person:

You want to check if two people are friends or not. Which graph implementation of your social network would have better runtime efficiency?

- [ ] Adjacency List
- [x] ~~Adjacency Matrix~~

If you were trying to check whether the i-th and j-th person are friends, you would just need to check whether adj[i][j] == 1, which is constant time. If you modeled it with an adjacency list, you would have to iterate through the whole list of either adj[i] or adj[j], which is average linear time.

First Person:

You want to check how many friends a specific person has. Which graph implementation of your social network would have better runtime efficiency?

- [x] ~~Adjacency List~~
- [ ] Adjacency Matrix

If you were trying to find the number of friends the i-th person has, you would just need to return the length of their list (len(adj[i])), which is constant time. If you modeled it with an adjacency matrix, you would have to find the sum of the i-th row/column (sum(adj[i]), which is linear time.